

SatQuery – EarthMind 4B Fine-Tuning Report

Progress from dataset preparation to the first complete fine-tuning run

1. Project Overview

SatQuery is being developed for Smart India Hackathon 2026, Problem Statement SIH26167. The objective is to build an AI system that can understand satellite images and answer natural-language questions about them. EarthMind 4B was selected as the base remote-sensing Vision-Language Model (VLM) for the first fine-tuning experiment.

2. Dataset Selection

Remote-sensing datasets containing satellite imagery and textual annotations were studied. BigEarthNet was one of the important sources considered because it contains Sentinel-1 and Sentinel-2 imagery with associated annotations. Because the complete dataset is very large, an approximately 10,000-sample subset was selected for the initial SatQuery experiment. The final selected dataset contains 9,977 samples and is stored in train_10k.parquet.

3. Dataset Information

Field	Purpose
ID	Unique sample identifier
s1_name	Sentinel-1 scene name
patch_id	Sentinel-2 scene name
input	Natural-language question
output	Expected answer
type	Question/answer type
category	Question category
split	Dataset split information
latitude / longitude	Geographic information
country	Country information
season	Season information
climate_zone	Climate information
s1_available / s2_available	Sensor availability

Example: Question: “Is any arable land and pastures adjoining regions in this scene?” Answer: “yes”. Type: binary. Category: adjacency.

4. Satellite Data

Each annotation is linked to actual satellite imagery. Sentinel-1 provides SAR information with VV and VH channels. Sentinel-2 provides optical/multispectral information with B01, B02, B03, B04, B05, B06, B07, B08, B09, B11, B12 and B8A. The parquet rows were mapped to the corresponding S1 and S2 scene folders and TIFF files.

5. Initial Image Preprocessing

Sentinel-1: VV, VH and VV–VH were combined to form a 3-channel representation. Robust normalization using the 2nd and 98th percentiles was applied.

Sentinel-2: B04, B03 and B02 were used as the first RGB representation. Bands were aligned/resampled where required.

Both modalities were converted to 3-channel images and resized to 448×448 before being sent to EarthMind.

6. EarthMind 4B and Fine-Tuning Approach

The pretrained EarthMind 4B model was loaded locally on an NVIDIA RTX A6000 with 48 GB VRAM. We are not training the 4B model from scratch. Instead, we are fine-tuning the pretrained EarthMind model using LoRA (Low-Rank Adaptation). The main language model and visual encoder were kept frozen while trainable LoRA components were updated.

7. Training Configuration

Parameter	Value
LoRA rank	128
LoRA alpha	256
LoRA dropout	0.05
Optimizer	AdamW
Learning rate	4×10^{-5}
Weight decay	0.05
Gradient clipping	1.0
Batch size	1
Gradient accumulation	4
Epochs	1

8. Image Tokens and Labels

EarthMind uses image tokens. The experiment used an image-start token, 256 image-context tokens, and an image-end token, followed by the natural-language question. Prompt tokens were masked with IGNORE_INDEX (-100), while answer tokens were used for the language-model loss.

9. Pipeline Validation

Before full training, the complete pipeline was tested step by step. EarthMind loading, S1/S2 preprocessing, image-token construction, tokenization, forward pass, loss calculation, backward propagation, LoRA gradients, AdamW updates, real SatQuery sample processing and the real DataLoader were all successfully verified.

10. Full Fine-Tuning Run

After the tests passed, the actual 9,977-sample SatQuery dataset was used for one complete epoch of LoRA fine-tuning.

Training statistic	Result
Samples processed	9,977
Epochs completed	1
Micro steps	9,977
Optimizer steps	2,495
GPU	NVIDIA RTX A6000 48 GB
Training time	1.82 hours
Average time/sample	≈ 0.65 seconds

11. Checkpoints

During training, checkpoints were saved at optimizer steps 500, 1000, 1500 and 2000. After the complete epoch finished at optimizer step 2495, the final model state was saved as satquery_earthmind_FINAL_step_2495.pt, approximately 2.75

GB in size.

12. Current Status

Stage	Status
Dataset selection / 9,977-sample subset	Completed
S1/S2 data mapping	Completed
Image preprocessing	Completed
EarthMind 4B setup	Completed
LoRA configuration	Completed
Tokenization	Completed
Forward/backward/optimizer tests	Completed
DataLoader	Completed
1-epoch fine-tuning	Completed
Final checkpoint	Saved
Validation/test evaluation	Next step

13. Next Steps

- Inspect the dataset split column and establish appropriate validation/test data.
- Load the final fine-tuned EarthMind checkpoint.
- Run predictions on held-out samples.
- Calculate accuracy, exact match and category-wise performance where appropriate.
- Compare original pretrained EarthMind with the fine-tuned model on the same evaluation data.
- Use the evaluation results to decide whether another epoch or training change is required.
- Later, investigate using the complete Sentinel-2 multispectral bands instead of only B04/B03/B02.

14. Summary

The first complete EarthMind fine-tuning experiment for SatQuery has been successfully completed. A 9,977-sample S1/S2 satellite question-answer dataset was connected to EarthMind 4B, the training pipeline was validated end-to-end, and one full epoch of LoRA fine-tuning was completed in approximately 1.82 hours. The final checkpoint has been saved. The next stage is systematic evaluation of the fine-tuned model.